

Face Entry Pro

Biometric Attendance & Campus Intelligence System

Prototype Design

A prototype is an early working model of a software system that is developed to demonstrate its design, features, and functionality before the final product is built. It helps developers and users evaluate the system, gather feedback, and identify improvements early in the development process.

The prototype of **FaceEntryPro** was developed to present the overall user interface, system workflow, and core functionalities before implementing the complete system. It provides a visual representation of how users interact with the application, allowing the design, navigation, and user experience to be evaluated and improved at an early stage of development.

The prototype includes the main interfaces of the system, such as user authentication, face enrollment, attendance management, dashboard, and administrative functions. It serves as a proof of concept that helps validate the system requirements and gather feedback before the final implementation.

Prototype Development Tool:

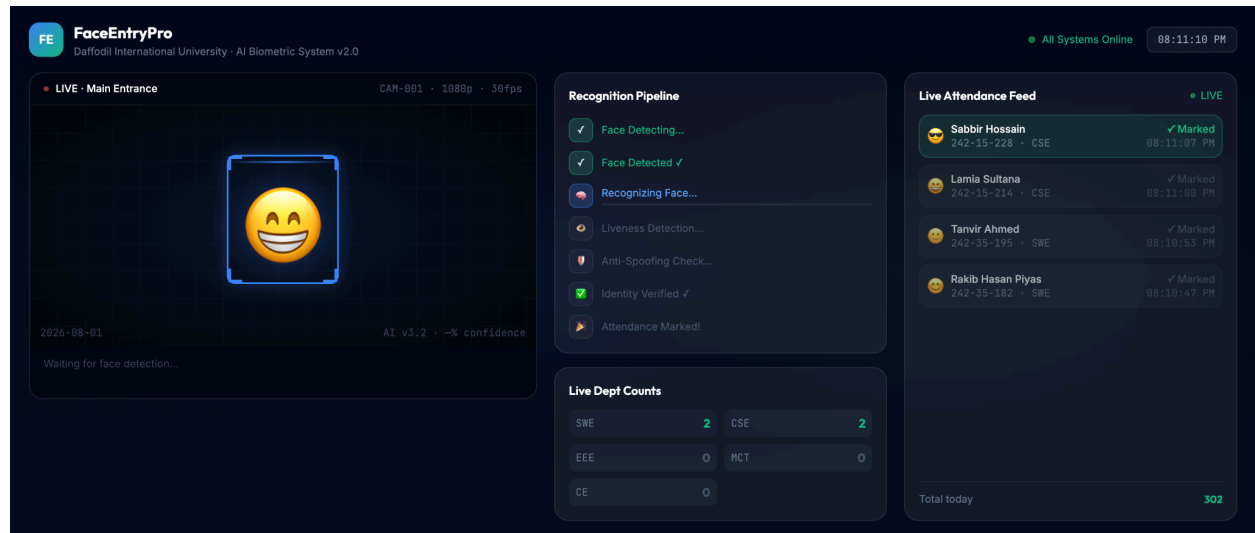
The prototype was designed using **Figma Make**, a web-based UI/UX design and prototyping tool. Figma Make was used to create interactive screens, design the application layout, and simulate the navigation flow between different interfaces, enabling stakeholders to visualize the proposed system effectively.

Prototype Link:

<https://www.figma.com/make/duqTBPQIi6fdBMFy5szE4D/Face-Entry-Pro?t=Vz5XCXd9jlBVgZB-20&fullscreen=1>

Note: The screenshots below provide a preview of the system's key functionalities. To experience the complete workflow and explore all features interactively, please visit the FaceEntryPro Figma Live Prototype.

1. Main Interface



The Live Attendance Dashboard is the main interface of the FaceEntryPro system for monitoring real-time attendance. It displays the live camera feed, recognition pipeline, attendance records, and department-wise statistics. The dashboard enables administrators to monitor face recognition, verify attendance, and check the system status through a simple and user-friendly interface.

Purpose:

Monitor real-time face recognition and attendance.

How to Use:

1. Open the Live Attendance Dashboard.
2. Ensure the system status shows "All Systems Online."
3. Position the student's face in front of the camera.
4. The system automatically performs face detection, recognition, liveness detection, and anti-spoofing verification.
5. After successful verification, attendance is marked automatically.
6. Check the Live Attendance Feed and Live Department Counts for attendance updates.

Dashboard Components:

- Live Camera Feed: Displays the real-time camera view.
- Recognition Pipeline: Shows each recognition step.
- Live Attendance Feed: Displays recently marked attendance.
- Live Department Counts: Shows department-wise attendance.

- **System Status:** Indicates whether the system is operating normally.

2. Login form

The screenshot displays the FaceEntryPro login interface. On the left, a dark blue panel titled 'Smart Attendance Powered by AI' lists features: 99.7% Recognition Accuracy, Real-time Processing, Anti-Spoofing Protection, and 256-bit Encryption. Below this, a 'Camera Network Status' section shows three camera locations (Main Entrance, Library, Lab-301) all marked as 'LIVE'. The right panel, titled 'Welcome back', prompts the user to sign in. It includes tabs for 'Student', 'Faculty', 'Parent', and 'Admin'. The 'Student' tab is selected. Below the tabs are input fields for 'Username / Student ID' (with a placeholder 'e.g. 242-35-182') and 'Password'. A 'Remember me' checkbox and a 'Forgot password?' link are present. A prominent blue 'Sign In' button is below the password field. An alternative 'Login with Face ID' button is shown below the 'Sign In' button, separated by an 'or' text. At the bottom, a link for 'New student? Create account' is visible. The footer contains the copyright notice: '© 2026 DIU Smart Campus System · FaceEntryPro v2.0'.

The Login Form is the entry point of the FaceEntryPro system, providing secure access for different types of users, including Students, Faculty, Parents, and Administrators. Users can sign in using their username or student ID and password, or authenticate through Face ID for faster and more secure access. The interface also displays the system's key features and the live status of connected campus cameras.

Purpose:

Provide secure user authentication to access the FaceEntryPro system.

How to Use

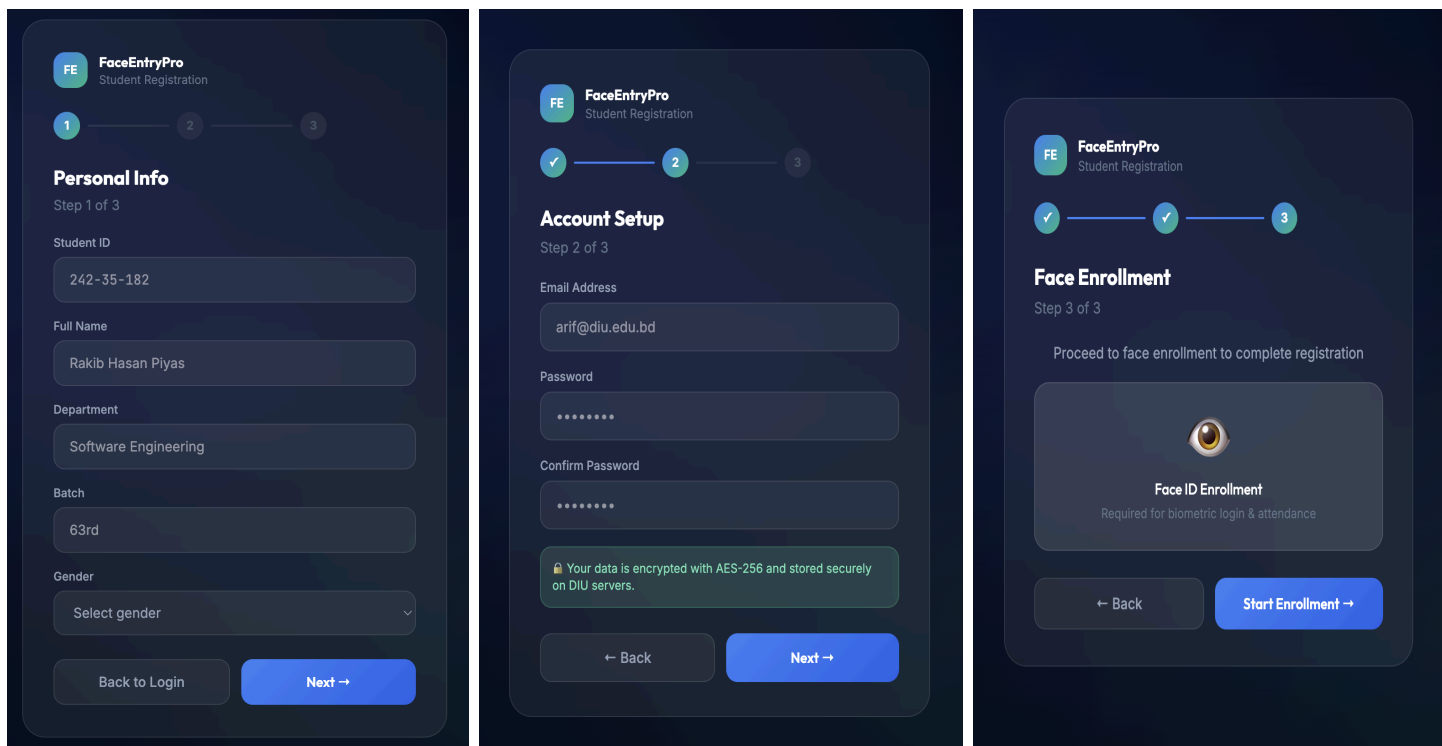
1. Select the appropriate user role (Student, Faculty, Parent, or Admin).
2. Enter your Username/Student ID and Password.
3. Click Sign In to access the dashboard.
4. Alternatively, click Login with Face ID for biometric authentication.

5. New users can select Create Account to register, and existing users can use Forgot Password if needed.

Login Form Components

- User Role Selection: Choose Student, Faculty, Parent, or Admin.
- Username/Student ID: Enter your login ID.
- Password: Enter your password securely.
- Sign In: Authenticates the user and grants system access.
- Login with Face ID: Enables biometric login using facial recognition.
- Create Account: Registers a new user.
- Forgot Password: Allows password recovery.
- Camera Network Status: Displays the status of connected campus cameras.

3. Register



The registration process consists of three steps:

- Step 1 of 3: Personal Info**
Fields include: Student ID (242-35-182), Full Name (Rakib Hasan Piyas), Department (Software Engineering), Batch (63rd), and Gender (Select gender). Navigation: Back to Login, Next →.
- Step 2 of 3: Account Setup**
Fields include: Email Address (arif@diu.edu.bd), Password, and Confirm Password. A security notice states: "Your data is encrypted with AES-256 and stored securely on DIU servers." Navigation: ← Back, Next →.
- Step 3 of 3: Face Enrollment**
Instruction: "Proceed to face enrollment to complete registration." A "Face ID Enrollment" section notes it is "Required for biometric login & attendance." Navigation: ← Back, Start Enrollment →.

The Registration Form allows new users to create a FaceEntryPro account through a simple three-step process. Users first provide their personal information, then create their account credentials, and finally complete Face Enrollment for biometric authentication.

Purpose:

Register a new user and enroll their face for biometric attendance.

How to Use

1. Enter your Personal Information (Student ID, Name, Department, Batch, and Gender).
2. Provide your Email Address and create a secure password.
3. Review the information and click Next to continue.
4. Complete the Face Enrollment by clicking Start Enrollment.
5. After successful enrollment, the registration process is completed.

Registration Form Components

- Personal Information: Collects student details.
- Account Setup: Creates login credentials.
- Face Enrollment: Registers the user's facial biometric data.
- Next / Back: Navigate between registration steps.
- Start Enrollment: Begins facial registration for biometric authentication.

4. Student Dashboard



FaceEntryPro is a modern, AI-powered face recognition attendance management system designed for students. It allows users to mark attendance using facial recognition, monitor attendance statistics, track daily class status, and view weekly and semester attendance reports through an interactive dashboard.

User Manual

1. Login

- Sign in using your student credentials.
- The dashboard opens after successful authentication.

2. Dashboard

The dashboard displays:

- Today's Status (Present/Absent)
- Attendance Rate
- Classes Today
- Total Classes Held
- Classes Attended
- Classes Missed
- Late Attendance Count
- Weekly attendance chart
- Semester attendance breakdown

3. Mark Attendance

- Click the "Mark Attendance" button.
- Allow camera access.
- Position your face in front of the camera.
- Wait for successful face verification.
- Your attendance status will update automatically.

4. View Attendance

- Open the Attendance menu from the left sidebar.
- Check your attendance history and daily records.

5. Profile

- Select Profile to view or update your personal information.

6. Analytics

- Open Analytics to monitor attendance trends and performance.

7. Notifications

- View important announcements and attendance-related alerts.

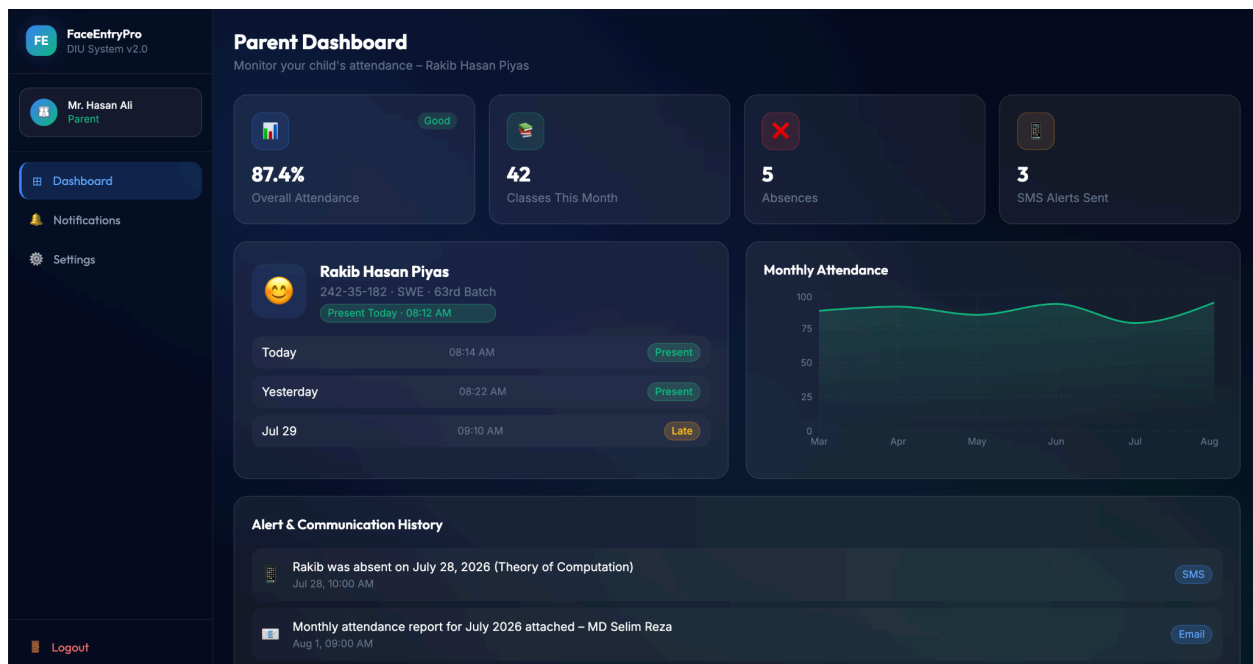
8. Settings

- Manage account preferences and system options.

9. Logout

- Click Logout to securely end your session.

5. Parent Dashboard



Parent Dashboard is a web-based attendance monitoring system that enables parents to track their child's attendance in real time. It provides an overview of attendance percentage, monthly class records, absence history, attendance trends, and

communication alerts, helping parents stay informed about their child's academic participation.

1. Login

- Log in using the parent's account credentials.
- The Parent Dashboard will open after successful authentication.

2. Dashboard Overview

The dashboard displays:

- Overall Attendance Percentage
- Classes This Month
- Total Absences
- SMS Alerts Sent
- Student's latest attendance status and check-in time
- Monthly attendance graph
- Alert and communication history

3. Monitor Attendance

- View your child's daily attendance status.
- Check attendance records for today and previous days.
- Review attendance trends using the monthly attendance chart.

4. View Notifications

- Open the Notifications menu to receive absence alerts, attendance updates, and important announcements.

5. Communication History

- Review all SMS and email notifications sent regarding attendance and monthly reports.

6. Settings

- Access Settings to manage account preferences and notification options.

7. Logout

- Click Logout to securely end your session.

6. Faculty Dashboard



Faculty Dashboard is a web-based attendance management system that enables faculty members to monitor student attendance in real time. It provides attendance statistics, department-wise performance, live attendance monitoring, monthly trends, and reporting tools to help instructors efficiently manage classroom attendance.

1. Login

- Log in using your faculty credentials.
- The Faculty Dashboard will appear after successful authentication.

2. Dashboard Overview

The dashboard displays:

- Total Students
- Students Present Today
- Students Absent Today
- Low Attendance Students (<75%)
- Department attendance statistics
- Monthly attendance trend chart
- Live attendance monitoring table

3. Live Monitor

- Click Live Monitor to track attendance as students check in using face recognition.
- View each student's attendance status in real time.

4. Attendance Reports

- Open Attendance Report or click Generate Report to create attendance reports for classes or departments.

5. Face Monitor

- Access Face Monitor to supervise the face recognition attendance process during class.

6. Analytics

- Use the Analytics section to view attendance trends, department comparisons, and student performance.

7. Student Management

- Review student attendance percentages and current status.
- Click View to see individual attendance details.
- Use Mark Manual when manual attendance correction is required.
- Click Send Alert to notify students with low attendance.

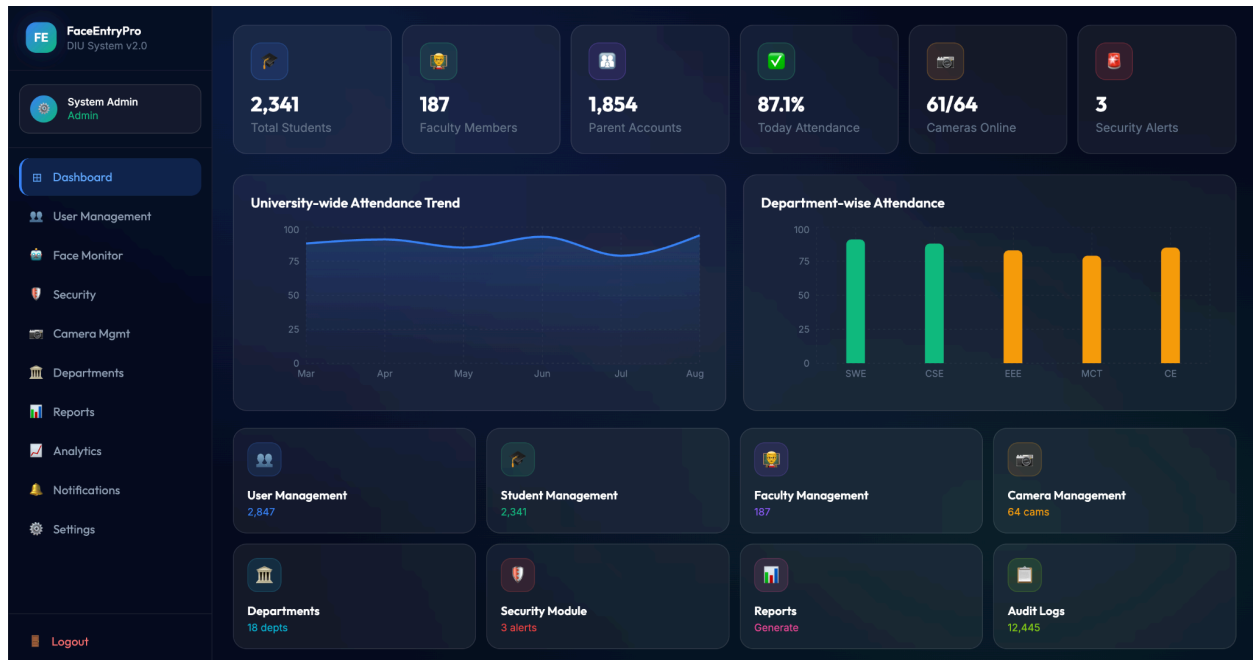
8. Notifications & Settings

- Check Notifications for system updates and alerts.
- Use Settings to manage account preferences.

9. Logout

- Click Logout to securely sign out of the system.

7. Admin Dashboard



Admin Dashboard is a centralized web-based management system that enables administrators to monitor and control the entire face recognition attendance platform. It provides real-time statistics, user management, department monitoring, security oversight, camera management, reporting, and analytics to ensure efficient and secure operation across the university.

1. Login

- Log in using your administrator credentials.
- The Admin Dashboard will load after successful authentication.

2. Dashboard Overview

The dashboard displays:

- Total Students
- Faculty Members
- Parent Accounts
- Today's Attendance Rate
- Cameras Online
- Security Alerts
- University-wide attendance trends

- Department-wise attendance statistics

3. User Management

- Open **User Management** to create, update, or remove student, faculty, and parent accounts.

4. Face Monitor

- Access **Face Monitor** to supervise live attendance sessions and monitor face recognition activities.

5. Security

- View security alerts, system events, and access logs to ensure platform security.

6. Camera Management

- Monitor camera status, configure devices, and troubleshoot offline cameras.

7. Departments

- Manage department information and monitor department-wise attendance performance.

8. Reports & Analytics

- Generate attendance reports and review university-wide attendance statistics and trends through the **Reports** and **Analytics** modules.

9. Notifications & Settings

- Check system notifications and configure application settings and administrator preferences.

10. Logout

- Click **Logout** to securely sign out of the system.

8. Attendance Reports (Faculty)



Attendance Reports is a reporting module that enables faculty members to generate, preview, and export attendance reports based on department, course, and date range. It provides attendance statistics, graphical analysis, and export options in PDF and Excel formats for easy documentation and record keeping.

1. Open Attendance Reports

- Select Attendance Report from the left navigation menu.

2. Set Report Filters

Choose the required filters:

- Department
- Course
- From Date
- To Date

3. Generate Report

- Click Generate Report to display the attendance report based on the selected filters.

4. Review Report

The report preview displays:

- Total Classes
- Average Attendance
- Total Students
- Students with Low Attendance
- Daily attendance chart showing Present and Absent students.

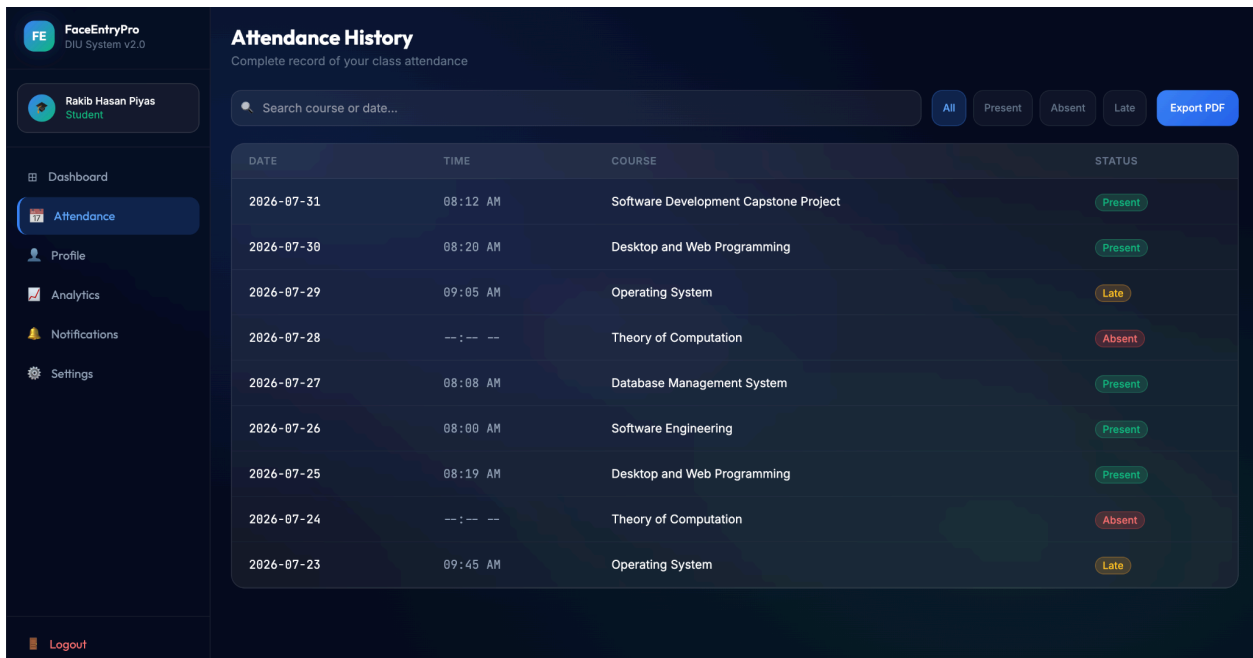
5. Export Report

- Click Export PDF to download the report as a PDF file.
- Click Export Excel to download the report in Excel format.

6. Update Filters

- Modify the filter options and regenerate the report to view different attendance records.

9. Attendance History (Student)



DATE	TIME	COURSE	STATUS
2026-07-31	08:12 AM	Software Development Capstone Project	Present
2026-07-30	08:20 AM	Desktop and Web Programming	Present
2026-07-29	09:05 AM	Operating System	Late
2026-07-28	--:--:--	Theory of Computation	Absent
2026-07-27	08:08 AM	Database Management System	Present
2026-07-26	08:00 AM	Software Engineering	Present
2026-07-25	08:19 AM	Desktop and Web Programming	Present
2026-07-24	--:--:--	Theory of Computation	Absent
2026-07-23	09:45 AM	Operating System	Late

Attendance History is a student attendance tracking module that provides a complete record of class attendance. Students can search attendance records, filter them by status (Present, Absent, or Late), and export their attendance history as a PDF for future reference.

1. Open Attendance History

- Select Attendance from the left navigation menu.

2. Search Records

- Use the Search bar to find attendance records by course name or date.

3. Filter Attendance

Choose one of the available filters:

- All – Displays all attendance records.
- Present – Shows only attended classes.
- Absent – Displays missed classes.
- Late – Shows classes where attendance was marked late.

4. View Attendance Details

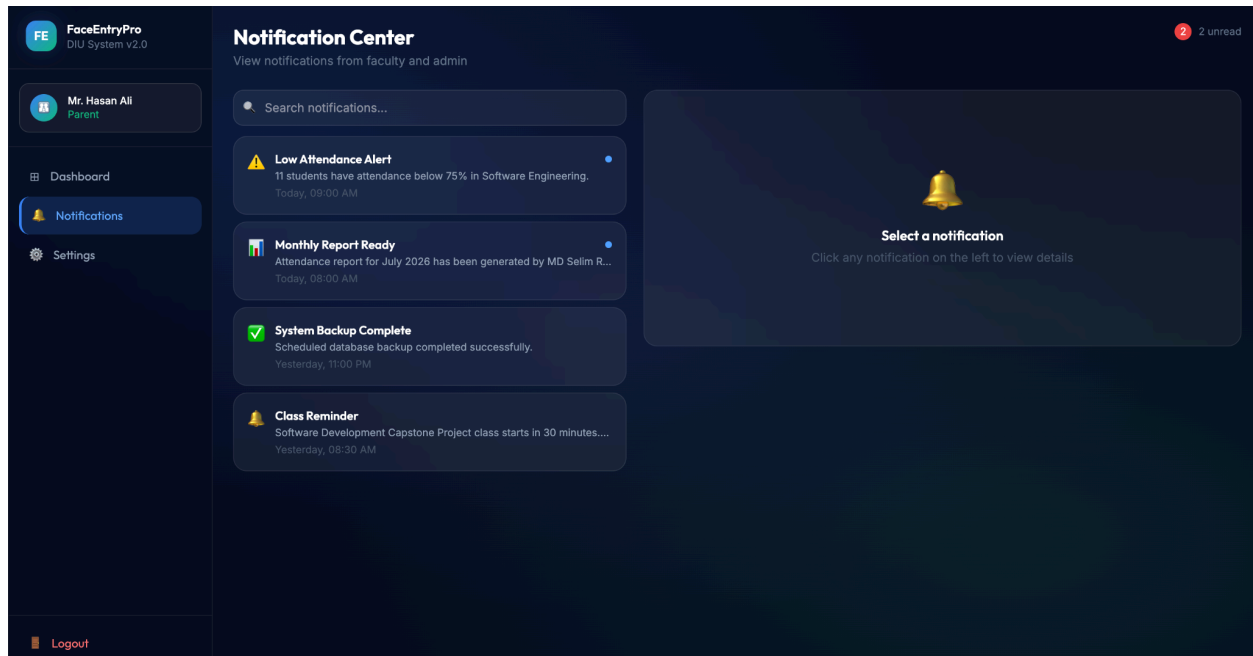
Each record includes:

- Date
- Attendance Time
- Course Name
- Attendance Status (Present, Absent, or Late)

5. Export Attendance

- Click Export PDF to download your attendance history as a PDF document.

10. Notification Center (Parent)



Notification Center is a centralized communication module that keeps users informed about attendance updates, low attendance warnings, report availability, class reminders, and important system announcements. It enables users to quickly access and review notifications from faculty and administrators.

1. Open Notification Center

- Select Notifications from the left navigation menu.

2. Search Notifications

- Use the Search bar to find notifications by title or keyword.

3. View Notifications

- Click on any notification from the list to display its details in the preview panel.
- Unread notifications are marked with an indicator.

4. Review Notification Types

The Notification Center provides alerts such as:

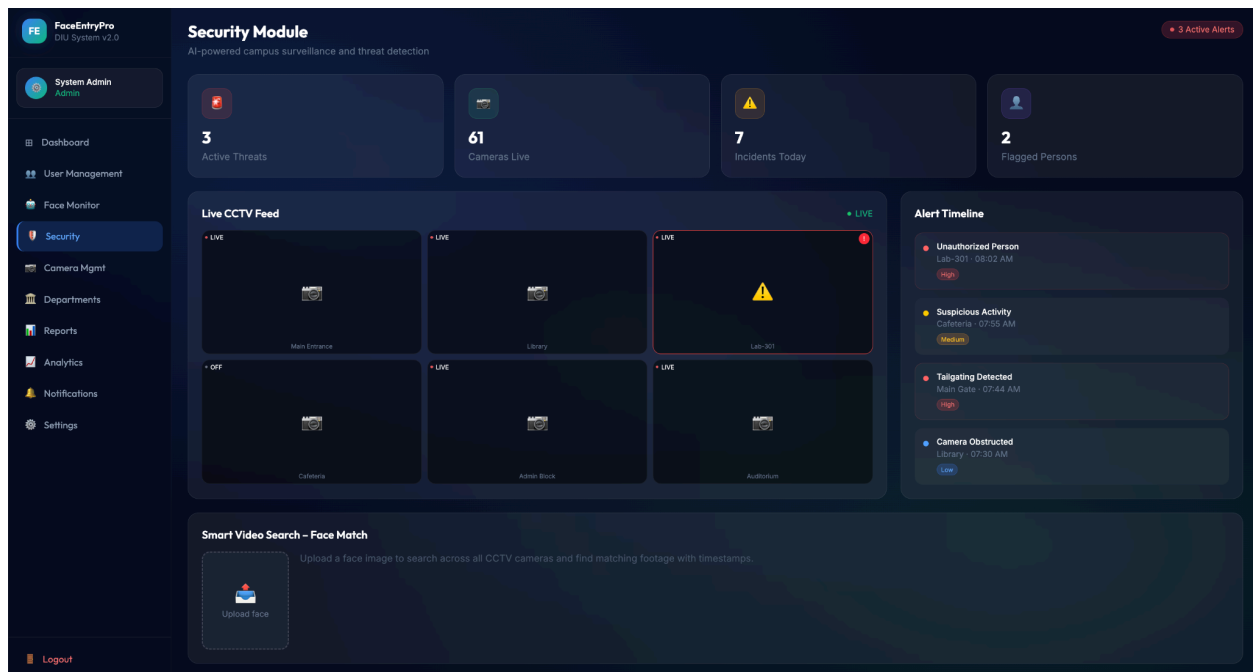
- Low Attendance Alerts
- Monthly Report Notifications

- System Backup Updates
- Class Reminders
- Other important announcements from faculty or administrators.

5. Monitor Unread Messages

- The unread notification counter at the top-right shows the number of notifications that require your attention.

11. Security Module (Admin : Suspicious activity detection and Smart video search)



Security Module is an AI-powered surveillance and security management system that monitors campus activities in real time. It provides live CCTV monitoring, threat detection, incident tracking, security alerts, and smart face search to help administrators maintain a safe and secure campus environment.

User Manual

1. Open Security Module

- Select Security from the left navigation menu.

2. Dashboard Overview

The dashboard displays:

- Active Threats
- Live Cameras
- Incidents Today
- Flagged Persons
- Current active security alerts

3. Monitor Live CCTV

- View live camera feeds from different campus locations.
- Check the status of each camera (Live or Offline).
- Monitor alerts highlighted on camera feeds.

4. Review Alert Timeline

- View detected security incidents in chronological order.
- Each alert includes:
 - Incident type
 - Location
 - Detection time
 - Severity level (High, Medium, or Low)

5. Smart Video Search

- Upload a face image using the Upload Face option.
- The system searches CCTV recordings for matching faces and displays matching footage with timestamps.

6. Respond to Alerts

- Review active threats and investigate suspicious activities.
- Take appropriate action based on the alert severity and incident details.

7. Logout

- Click Logout to securely exit the system.
-